

Ph.D Student at KAIST AI

Contact Information

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Interest

My long-term goal is to build models that can reason beyond human data distribution. To this end, I study the learning signals, data structures, and evaluation methods that enable models to self-train and generalize systematically in multi-step reasoning. Currently, I am interested in iterative RL post-training and agentic systems, particularly autonomous research agents.

Experience

Research Intern

AI Companion Team, KRAFTON AI, Seoul, Republic of Korea May 2025 – Nov 2025
 - Dept. Lead: Kangwook Lee
 - Topic: Developing Game-Play Agent (Co-playable Character) for PUBG (w. NVIDIA)
 (Link: <https://www.youtube.com/watch?v=wEKUSMqrbzQ>)

Research Intern

LK Lab, KAIST, Remote, Republic of Korea
 - Mentor: Seonghyeon Ye
 - Topic: Prompting Mechanisms of LLMs, Fine-grained Evaluation Metric

Research Intern

Naver CLOVA, Healthcare AI, Seongnam, Republic of Korea
 - Mentor: Yera Choi
 - Topic: EHR Summarization and embedding

Mar 2022 – Aug 2022

Research Intern

KCVL Lab, KAIST, Daejeon, Republic of Korea
 - Mentor: Hangil Park
 - Topic: 2D Classification and Detection on imbalanced dataset

Jun 2021 – Sep 2021

Education

KAIST

Ph.D., Graduate School of AI
Advisor : Seongjoon Oh

KAIST

M.S., Graduate School of AI
Advisor : Minjoon Seo

KAIST

B.S., School of Computing
Advisor : Tae-Kyun Kim
GPA: 3.96 / 4.3

Languages

- Korean	Native
- English	Fluent (TOEFL 113)
- Spanish	Fluent (DELE C1)

Honors & Awards

- Young-Han Kim Global Leadership Scholarship Awardee (2022)
- Qualcomm-KAIST Innovation Award Hackathon Winner (2023)

Reviewer Experience

- ICLR (2026)
- ICML (2026, Gold Reviewer [Top 25%])
- NeurIPS (2026)
- TMLR (2026)

Publications

[1] Hwang, H., & Park, Y. (2026). *Intrinsic Task Symmetry Drives Generalization in Algorithmic Tasks*. **(ICML 2026)**

[2] Hwang, H., Jeon, B., Kim, S., Kim, J., Chang, H., Yang, S., Won, S., Lee, D., Ahn, Y., ... (2025). *Let's Predict Sentence by Sentence*. arXiv preprint arXiv:2505.22202. **(COLM RAM 2 Workshop [Oral])**

[3] Won, Y., Lee, H., Hwang, H., & Seo, M. (2025). *Differential Information: An Information-Theoretic Perspective on Preference Optimization*. arXiv preprint arXiv:2505.23761. **(Under Review)**

[4] Chang, H., Park, J., Cho, H., Yang, S., Ko, M., Hwang, H., Won, S., Lee, D., Ahn, Y., ... (2025). *The Coverage Principle: A Framework for Understanding Compositional Generalization*. arXiv preprint arXiv:2505.20278. **(ICLR 2026)**

[5] Lee, S., Kim, S., Seo, M., Jo, Y., Go, D., Hwang, H., Park, J., Yue, X., Welleck, S., ... (2025). *The CoT Encyclopedia: Analyzing, Predicting, and Controlling how a Reasoning Model will Think*. arXiv preprint arXiv:2505.10185. **(ICLR 2026)**

[6] Kim, J., Lee, H., Cho, H., Jang, J., Hwang, H., Won, S., Ahn, Y., Lee, D., & Seo, M. (2024). *Knowledge Entropy Decay during Language Model Pretraining Hinders New Knowledge Acquisition*. **(ICLR 2025 [Oral] & AAAI Workshop KnowFM 2025 [Best Paper])**

[6] Kim, S., Suk, J., Cho, J. Y., Longpre, S., Kim, C., Yoon, D., Son, G., Cho, Y., & Hwang, H. (2024). *The BiGGen Bench: A Principled Benchmark for Fine-grained Evaluation of Language Models with Language Models*. **(NAACL 2025 [BEST PAPER])**

[7] Hwang, H., Kim, D., Kim, S., Ye, S., & Seo, M. (2024). *Self-Explore to Avoid the Pit: Improving the Reasoning Capabilities of Language Models with Fine-grained Rewards*. **(EMNLP Findings (2024) & ACL Workshop NLRSE 2024 [Oral])**

[8] Ye, S., Hwang, H., Yang, S., Yun, H., Kim, Y., & Seo, M. (2023). *Investigating the Effectiveness of Task-Agnostic Prefix Prompt for Instruction Following*. **(AAAI 2024)**

[9] Ye, S., Kim, D., Kim, S., Hwang, H., Kim, S., Jo, Y., Thorne, J., Kim, J., & Seo, M. (2023). *FLASK: Fine-grained language model evaluation based on alignment skill sets*. **(ICLR 2024 [Spotlight])**

[10] Hwang, H., Kim, S., Park, W. J., Seo, J., Ko, K., & Yeo, H. (2022). *Vision transformer equipped with neural resizer on facial expression recognition task*. **(ICASSP 2022)**